

Question_8:

Code:

```
1  /*
2  QUESTION_8:
3  Fold a linked list.
4  The folding order is: First node, Last node,
5  |         |         |         |         |         Second node, Second-last node,
6  |         |         |         |         |         Third node, Third-last node
7  */
8  #include <iostream>
9  #include <vector>
10 using namespace std;
11
12 struct node {
13     int data;
14     node *next;
15 };
16 void create(node *&head, int x) {
17     node *p = new node{x, NULL};
18
19     if (!head)
20         head = p;
21     else {
22         node *t = head;
23
24         while (t->next)
25             t = t->next;
26         t->next = p;
27     }
28 }
29 void print(node *head) {
30     while (head) {
31         cout << head->data << " ";
32         head = head->next;
33     }
34     cout << endl;
35 }
36 int main() {
37     int n, x;
38     cin >> n;
```

```
39
40     node *head = NULL;
41     for (int i = 0; i < n; i++) {
42         cin >> x;
43         create(head, x);
44     }
45     vector<int> a;
46
47     for (node *p = head; p; p = p->next)
48         a.push_back(p->data);
49     cout << "Link list data:";
50     print(head);
51
52     cout << "Link list data after fold:";
53     int left = 0;
54     int right = a.size() - 1;
55
56     while (left <= right) {
57         cout << a[left++] << " ";
58
59         if (left <= right)
60             cout << a[right--] << " ";
61     }
62     return 0;
63 }
```

Output:

Test Case_1:

```
16
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
Link list data:1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
Link list data after fold:1 17 3 98 4 32 7 56 9 24 11 23 13 19 14 18
PS C:\Users\Prabin Yadav\Documents\III SEM\DSA\DSA Lab Activity Solved\
```

Test Case_2:

```
15
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
Link list data:0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
Link list data after fold:0 2 1 1 2 0 0 2 1 2 1 0 1 2 0
PS C:\Users\Prabin Yadav\Documents\III SEM\DSA\DSA Lab Ac
```